

leaves a void where respective components are to be placed, and pauses at predetermined places where components need to be embedded. You can remove the print platform or print bed from the device, insert the component and place it back; and the printing process continues from where it was paused. Kinematic coupling and magnets are used to precisely fix the platform back in its exact place, when it is returned after manual embedding of components.

Only for rigid electronics

Voxel8 is not without its drawbacks. One issue we find with it is that it supports just two materials, namely, PLA plastic and conductive silver ink. However, the team has hinted at more functionalities and materials being introduced in the future. This promises a lot of opportunities for electronics enthusiasts.

Another limitation of this printer is that the conductive inks used in the existing methods are not very conductive. These offer mediocre conductivity and are suited only for simple, low-power applications.

The third drawback is that it can be used to print only rigid electronics and not flexible ones.

Flexible electronics is becoming very popular now, especially in medical, entertainment and wearable domains. It calls for electronics printing methods that are fast, cheap and easy, and can produce circuits in bulk. This creates the need for better technology that can also print electronics circuitry in flexible surfaces, like plastic sheets or even clothing materials.

The solution—inkjet printing

By using inkjet printing to produce electronic circuits, we can print flexible and stretchable conductors onto anything, including elastic materials and fabrics. Here we look at how two teams are using this technology:

The Purdue method. Folks at Purdue University have come up with the concept of using regular

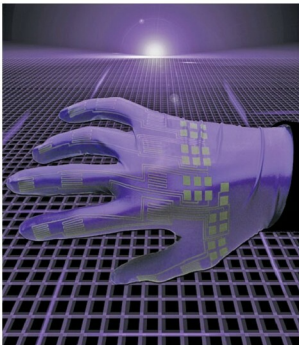


Fig. 3: Inkjet for printing flexible electronics (Image courtesy: Purdue University)

Specifications of Voxel8 3D Printer

Brand	Voxel8
Materials	PLA Conductive ink (exclusive)
XY resolution	15 microns
Layer resolution	200 microns
Other features	10.9cm (4.3-inch) touchscreen Wi-Fi and USB connectivity
Cost	US\$ 9000
Availability	Pre-order; ships in late 2015

inkjet printers for printing flexible and stretchable electronics in mass quantities. A research paper on the method titled 'Mechanically Sintered Gallium-Indium Nanoparticles' has been published by Rebecca Kramer and her team in *Advanced Materials* journal, dated April 17, 2015.

The operation. In its native form, liquid metal cannot be used for inkjet printing. To overcome this, the team has used mechanically-sintered gallium-indium nanoparticles, where liquid metal is turned into nanoparticles by immersing the alloy into a solvent such as ethanol and then subjecting the mixture to ultrasound to disperse the liquid metal in it. This method

makes the liquid metal small enough to pass through an inkjet nozzle.

Once printing is done, the solvent evaporates and leaves behind nanoparticles and an oxidised gallium (Ga_2O_3) coating that is used to keep the liquid metal electrically inert during the process. This thin skin can be easily removed by stamping or dragging something across the surface, leaving a conducting wire on the material on which the design is printed. This also makes it possible to activate only selected portions, and

even modify or upgrade these later, depending on the particular design.

No heating required. Unlike other graphene based printing techniques, Purdue method does not require the printed material to be heated to very high temperatures to make it conductive. This simplicity makes the method more accessible to circuit designers.

Stretchable. Since these are made from liquid metal and can be printed on any material, including plastic sheets, papers or even fabrics, the circuits can be stretched, folded and squeezed without breaking. This can be utilised for creating wearable devices that do not restrict motion and soft machines (like robots) that can squeeze through small spaces.

The Cartesian method. Australian start-up Cartesian Co. started a Kickstarter campaign in November 2013 for developing Argentum (previously EX¹), which is supposedly the world's first desktop electronics printer. The project got funded and they started shipping their first batch in August 2014.

Similar to the previous method, Argentum lets you print both flex-